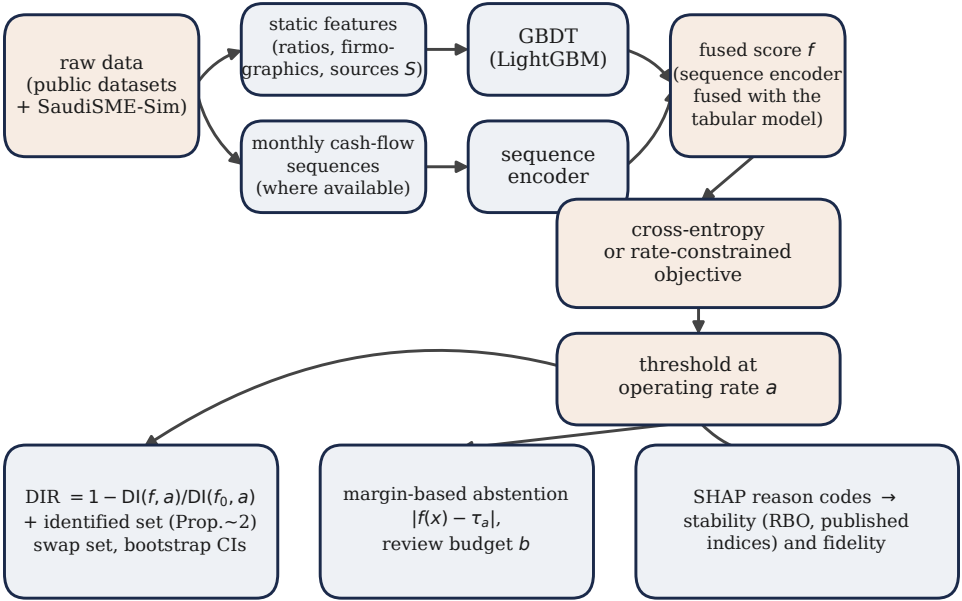
raw data
(public datasets
+ SaudiSME-Sim)
static features
(ratios, firmo-
graphics, sources S)
monthly cash-flow
sequences
(where available)
GBDT
(LightGBM)
sequence
encoder
fused score f
(sequence encoder
fused with the
tabular model)
cross-entropy
or rate-constrained
objective
threshold at
operating rate a
DIR $= 1 - \mathrm{DI}(f, a)/\mathrm{DI}(f_0, a)$
+ identified set (Prop.~2)
swap set, bootstrap CIs
margin-based abstention
$|f(x) - \tau_a|$,
review budget b
SHAP reason codes $\rightarrow$
stability (RBO, published
indices) and fidelity